

SOMALI INTERNATIONAL UNIVERSITY

Full-Stack Web Development I

Week 1 — The Developer Mindset, How the Web Works,
and Skills First, AI Second

Semester Five | Year Three | 2 Credit Hours

Where this course takes you

32

teaching weeks

across Semester 5 and Semester 6

4

competency levels

Understand, Build, Engineer, Deliver

1

goal

ship real software you can explain

Level 1 Understand → Level 2 Build → Level 3 Engineer → Level 4 Deliver

By the end of Semester 5 you build a complete full-stack application on your own. By the end of Semester 6 your team delivers, secures, tests, documents, deploys and defends a real system.

What we cover today

1

The software development mindset

Coding is not engineering. What developers really do.

2

Problem decomposition

How to break a large problem into solvable pieces.

3

How the Web works

DNS, client-server, request and response.

4

HTTP and HTTPS

Methods, status codes, headers, statelessness.

5

The map of the full stack

Frontend, backend, database, API.

6

Skills First, AI Second

The course philosophy and the AI usage policy.

PART ONE

The Software Development Mindset

Anyone can write code. Engineers decide what to build, why, and how it will survive contact with reality.

Coding is not the same as engineering

Coding

Writing instructions a computer can run

- Focus on making it work now
- One person, one file, one machine
- Success = the output appears
- Bugs are surprises
- Finished when it runs

Engineering

Building software that other people depend on

- Focus on making it work correctly and keep working
- Teams, repositories, environments, users
- Success = correct, secure, maintainable, understood
- Bugs are expected and hunted systematically
- Finished when it is tested, documented and deployed

This course grades engineering, not only working output. That is why every project ends with a defense.

What a developer actually spends time on

R

Reading

Reading existing code and documentation is most of the job.

T

Thinking

Choosing an approach before typing saves hours of rework.

D

Debugging

Finding out why reality disagrees with your expectation.

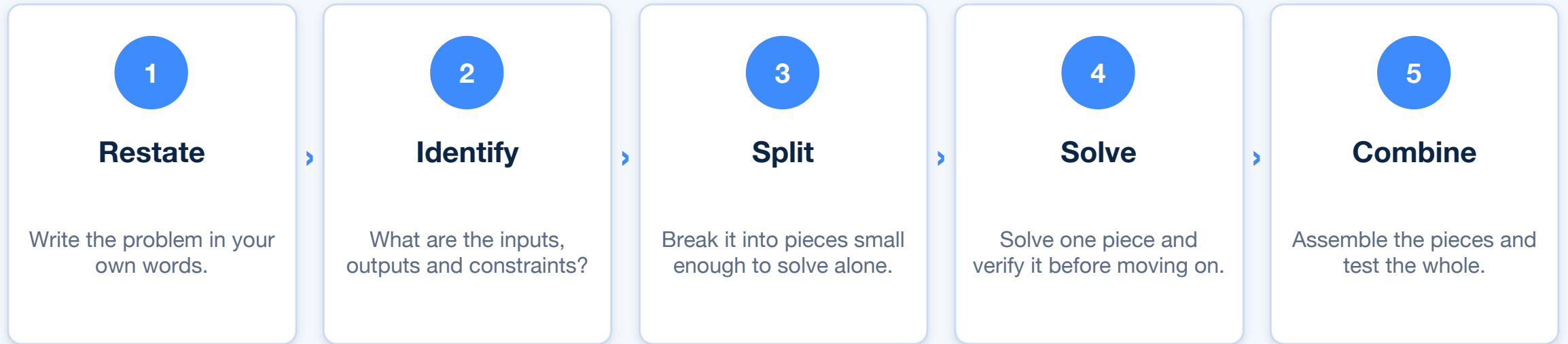
C

Communicating

Requirements, reviews, commit messages, documentation.

Typing code is the smallest part of the work — and it is the part AI is best at.

Problem decomposition: the core skill



Example: “Build a student attendance system” is not a task. “Store one attendance record for one student on one date” is a task.

PART TWO

How the Web Works

Before you build for the Web, you must be able to explain what happens when someone opens your page.

What happens when you press Enter

1

You type a URL

The browser parses the address into protocol, host and path.

2

DNS lookup

The domain name is translated into an IP address.

3

Connection

The browser opens a connection to that server, secured by TLS for HTTPS.

4

HTTP request

The browser sends a method, a path, and headers.

5

Server work

The server runs code, may query a database, and builds a response.

6

HTTP response

A status code, headers, and a body travel back.

7

Rendering

The browser builds the page and runs JavaScript.

Client and server: two machines, one conversation

CLIENT

The browser

- Runs on the user's device
- Requests resources
- Renders HTML and CSS
- Executes JavaScript
- Cannot be trusted

request →

← response

SERVER

The application

- Runs where you control it
- Applies business rules
- Talks to the database
- Enforces security
- Is the source of truth

Rule to remember: never trust the client. Every rule that matters is enforced again on the server.

HTTP: the language of the Web

GET

Read a resource

POST

Create a resource

PUT / PATCH

Replace or update

DELETE

Remove a resource

Every request carries

- A method and a path
- Headers (who is asking, what format)
- Sometimes a body (the data being sent)

HTTP is stateless

The server does not remember you between requests. Every request must carry whatever the server needs to know — which is why tokens and cookies exist. We build this in Semester 6.

Status codes tell you what happened

2xx

Success

200 OK · 201 Created · 204 No Content

3xx

Redirection

301 Moved · 304 Not Modified

4xx

You made a mistake

400 Bad Request · 401 Unauthorized · 403 Forbidden · 404 Not Found

5xx

The server made a mistake

500 Internal Server Error · 503 Unavailable

A 404 is not a broken server. It is the server telling you clearly that the thing you asked for does not exist.

HTTP and HTTPS

HTTP

Plain text. Anyone between the browser and the server can read and change the traffic — passwords included.

HTTPS

The same HTTP, wrapped in TLS encryption. The traffic is private and its integrity is protected in transit.

What HTTPS does not do

It does not make your application secure. It protects data while it travels. Weak passwords, missing validation and SQL injection are still your responsibility — that is Semester 6, Week 12.

The map of the full stack

Frontend

What the user sees and touches

HTML · CSS · Tailwind · JavaScript · React

Weeks 3–12

API layer

The contract between the two sides

HTTP · REST · JSON · Express routes

Weeks 9, 13–14

Backend

Rules, decisions, and security

Node.js · Express · controllers · services

Weeks 13–14

Database

Memory that survives a restart

SQL · PostgreSQL · schema design

Weeks 15–16

We learn each layer separately, then connect them. Fundamentals before abstractions: CSS before Tailwind, JavaScript before React, HTTP before REST, SQL before any ORM.

PART THREE — THE MOST IMPORTANT PART

Skills First, AI Second

Learn software development first. Use AI to multiply your productivity, not to replace your understanding.

Two ways to use AI — only one produces engineers

The trap

Problem → Ask AI → Copy → Paste → Submit

- The code may work today
- You cannot debug it tomorrow
- You cannot extend it when requirements change
- You cannot explain it in an interview
- You learned nothing

The professional way

Think → Understand → Design → Code → Test → Debug →

Use AI → Verify → Improve → Deliver

- You design the solution, AI accelerates it
- You read and test everything AI writes
- You reject what you do not understand
- You stay the engineer, not the passenger

AI is a power tool. In trained hands it is productivity. In untrained hands it is a way to stay a beginner forever.

THREE PHASES

How AI is introduced in this course

Phase A

Weeks 1–8

AI not permitted for graded work

Build real syntax fluency, problem-solving and debugging ability in HTML, CSS and JavaScript.

Phase B

Weeks 9–16

Supervised use with declaration

Learn to prompt well, verify output, and reject a wrong AI answer with confidence.

Phase C

Semester 6

Professional use, fully logged

Work at professional speed with full ownership and full understanding.

From Week 9, every project includes an AI-USAGE.md file: what tool you used, where you used it, and how you verified it.

THE GOLDEN RULE OF THIS COURSE

If you cannot explain it, you cannot submit it.

In every practical exam and defense, the examiner may open your repository and point at any component, function, API endpoint, SQL query, or database relationship — and ask you to explain it, then change it, live, with no AI.

Working code that you cannot explain scores zero.

How you will be assessed

20%

Assignments & Practices

30%

Midterm Examination

50%

Final Project

What counts as work

- Everything is submitted through GitHub
- Work not in your repository is not graded
- Commit steadily, not once before the deadline
- Four projects build on one another

Your weekly time

- 2 hours in class: concepts, live coding, checkpoints
- 3–4 hours on the weekly lab
- 1.5–2 hours preparing before class
- 2–3 hours on project work

This week's work

Lab 1 — Trace a real website

- 1 Open any website and open the Network tab in DevTools.
- 2 Record the URL, method, status code, response type and time for five requests.
- 3 Draw the client–server diagram for that website.
- 4 Write one page in your own words: “What happens when I type a URL and press Enter?”
- 5 Push everything to your course repository before the deadline.

Home task

Sign the AI Usage Policy acknowledgement and write your personal learning contract for the 32 weeks.

Reminder

No AI for this lab. Weeks 1–8 are how you earn the right to use it well later.

NEXT WEEK

Week 2 — Your professional toolkit

VS Code

Set up like a professional

Terminal

Move without a mouse

DevTools

See what the browser sees

Git & GitHub

Never lose work again

Come with your laptop charged, Node.js downloaded, and a GitHub account created.

Questions?